Quantum Mechanics Derived from Cost Minimization: The Forcing Chain from a Single Primitive to Full Quantum Theory

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

We derive the full structure of quantum mechanics from a single primitive: the Recognition Composition Law (RCL). A nine-level forcing chain (T0–T8) produces, in succession: logic (T0), existence over non-existence (T1), discreteness (T2), double-entry bookkeeping (T3), the recognition operator $\widehat{R}$ (T4), the unique cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ (T5), the golden ratio $\varphi = (1 + \sqrt{5})/2$ (T6), the 8-tick epoch (T7), and spatial dimension $D = 3$ (T8). From these forced structures, quantum mechanics emerges: complex Hilbert space (from the 8-tick half-period $e^{i\pi} = -1$), the Born rule $P = |\psi|^2$ (from J-cost phase invariance), entanglement (from shared ledger constraints), commutation relations $[\hat{x}, \hat{p}] = i\hbar$ (from double-entry non-commutativity), and definite measurement outcomes (from the discrete ledger update at $C \geq 1$). No measurement postulate, no Hilbert-space axiom, no Born rule assumption, and no free parameters are required.

1 Introduction

Quantum mechanics has a structure that works with extraordinary precision, yet its foundations remain disputed [2, 3]. Five foundational questions persist:

1. Why the Born rule $P = |\psi|^2$?
2. Why complex Hilbert space (not real or quaternionic)?
3. What is measurement, ontologically?
4. What is entanglement?
5. Why $[\hat{x}, \hat{p}] = i\hbar$?

Hardy [2] and Chiribella *et al.* [3] have reconstructed QM from information-theoretic axioms, but such reconstructions leave open: why *these* axioms? What forces them?

Recognition Science [6] provides a complete answer: quantum mechanics is *forced* by cost minimization from a single primitive—the Recognition Composition Law.

2 The Single Primitive

Definition 2.1 (Recognition Composition Law)**.** The cost of composing recognition events satisfies a symmetric, normalized defect measure. For a cost function F on positive ratios, the RCL states:

$$F(xy) + F(x/y) = 2F(x)F(y) + 2F(x) + 2F(y). \tag{1}$$

Together with three constraints:

(A1) **Normalization**: $F(1) = 0$.

(A2) **Symmetry**: $F(x) = F(1/x)$ for all $x > 0$.

(A3) **Calibration**: $F''(1) = 1$.

Theorem 2.2 (Uniqueness of the Cost Functional)**.** *The unique continuous function satisfying* (1) *and (A1)–(A3) is*

$$J(x) = \frac{1}{2}\left(x + x^{-1}\right) - 1. \tag{2}$$

Proof sketch. Setting $f(t) = F(e^t) + 1$ and substituting $x = e^s$, $y = e^t$ in (1) gives the d'Alembert equation $f(s+t) + f(s-t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ (from A1) and $f''(0) = 1$ (from A3) is $f(t) = \cosh t$, giving $F(x) = \frac{1}{2}(x + x^{-1}) - 1$. Symmetry (A2) is then automatically satisfied. □

3 The Forcing Chain T0–T8

Each level is *forced*, not postulated.

3.1 T0: Logic from Cost

Remark 3.1 (Logic from Cost — Domain Observation)*.* The domain of J is $(0, \infty)$. For $x \leq 0$, x^{-1} is undefined or negative, giving no valid cost assignment. In this sense, the domain of J is identified with the space of consistent states. This is a structural identification: consistent configurations are those with well-defined J-cost, and contradictions (states with $x \leq 0$) fall outside the domain. The RS claim that this derives "logic from cost" means: the logical constraint that states be consistent is equivalent to the requirement that J-cost be finite.

3.2 T1: Existence Over Non-Existence

Proposition 3.2 (Meta-Principle — Structural Argument)**.** *In the RS identification of $x \to 0^+$ with non-existence: $J(x) \to \infty$ (infinite cost for the zero-ratio limit). Only $x = 1$ has zero cost. Existence ($x = 1$) is structurally preferred over non-existence ($x \to 0^+$) because its cost is minimal.*

Proof. $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. As $x \to 0^+$, $x^{-1} \to +\infty$, so $J(x) \to +\infty$. At $x = 1$: $J(1) = 0$. Existence is cheaper than non-existence. □

3.3 T2: Discreteness

Proposition 3.3 (Discreteness Forced — Structural Argument)**.** *Continuous configurations converge to the unique minimum $x = 1$ under J-gradient flow. In a system with self-similar composition, the φ-ladder $\{\varphi^r : r \in \mathbb{Z}\}$ is the unique discrete set closed under both $x \mapsto \varphi x$ and $x \mapsto x^{-1}$.*

Structural argument. Under J-gradient flow, any $x \neq 1$ is driven toward $x = 1$ (the unique global minimum). The self-similar composition law $\varphi^2 = \varphi + 1$ (from T6) identifies the natural discrete scale as the φ-ladder. Configurations off the ladder are unstable under J-cost minimization.

Circular dependency note: The identification of the φ-ladder here relies on the golden ratio φ from T6, which formally appears later in the forcing chain. The forcing chain is therefore not strictly sequential at this step: T2 (discreteness) uses the output of T6 (φ). The correct interpretation is that T2 and T6 are mutually reinforcing: self-similar composition forces discreteness, and the unique ratio satisfying self-similar composition is φ. □

3.4 T3: Ledger (Double-Entry Bookkeeping)

Proposition 3.4 (Ledger Structure — Structural Identification)**.** *The symmetry $J(x) = J(1/x)$ motivates double-entry bookkeeping: every recognition event x has a matching inverse x^{-1} with equal cost, providing the RS origin of conservation laws.*

Proof. $J(x) = \frac{1}{2}(x + x^{-1}) - 1 = \frac{1}{2}(x^{-1} + x) - 1 = J(x^{-1})$. The cost cannot distinguish x from x^{-1}; both must be recorded. □

3.5 T4: Recognition Operator

Proposition 3.5 (Recognition Operator — Structural Identification)**.** *A discrete operator $\widehat{R}$ is identified as the fundamental evolution operator: it extracts observables from ledger states, enforces J-minimization, and maintains ledger consistency. $\widehat{R}$ replaces the Hamiltonian $\widehat{H}$.*

Structural argument. In the discrete ledger (T2), a physical quantity is observable only when it corresponds to a stabilized recognition event—a ledger entry with definite J-cost. The map from ledger states to observables must: (i) satisfy J-minimization (the cost is non-increasing under the dynamics), (ii) preserve the ledger double-entry structure (T3), and (iii) be realizable in the $2^D = 8$-tick period (T7). These three conditions uniquely specify a linear discrete-time operator on the Hilbert space of ledger states, which we call $\widehat{R}$. $\widehat{R}$ is a discrete analog of the time-evolution operator $e^{-i\widehat{H}\tau_0/\hbar}$, but derived from the J-cost structure rather than postulated. □

Dynamics are given by the discrete update:

$$s(t + 8\tau_0) = \widehat{R}\big(s(t)\big), \tag{3}$$

minimizing J-cost rather than energy.

3.6 T5: Unique J

The cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ is uniquely determined by the RCL plus normalization and calibration (Theorem 2.2).

3.7 T6: The Golden Ratio φ

Theorem 3.6 (φ Forced)**.** *Self-similar nesting in the discrete ledger requires $x^2 = x + 1$. The unique positive solution is $\varphi = (1 + \sqrt{5})/2$.*

Proof. The self-similar condition requires that the ratio between successive levels of the hierarchy satisfies $x = 1 + 1/x$, i.e., $x^2 = x + 1$. Solving: $x = (1 \pm \sqrt{5})/2$. The positive root is φ. □

3.8 T7: The 8-Tick Epoch

Proposition 3.7 (8-Tick Epoch — Structural Argument)**.** *The linking structure of the discrete ledger is identified as requiring a Hamiltonian cycle on the D-cube graph. The minimal Hamiltonian cycle length on Q_D (which has 2^D vertices) is 2^D.*

Structural argument. The discrete ledger in D spatial dimensions has 2^D states (one per sign pattern of the D spatial directions). The recognition cycle must visit all 2^D states to ensure complete ledger updating—a Hamiltonian cycle on the D-cube graph Q_D. The D-cube graph is bipartite (vertices alternately colored black and white by parity); the shortest Hamiltonian cycle on Q_D has length 2^D (it visits all vertices once and returns). For $D = 3$: $2^3 = 8$ ticks. □

Remark 3.8. T7 and T8 are mutually reinforcing: the ledger-linking structure forces $n = 8$ as the minimal period (from the unique non-trivial linking in $D = 3$), while $n = 8$ forces $D = 3$ via $2^D = 8$ (uniquely). The correct causal direction is: the linking structure of the recognition ledger forces the minimal period $n = 8$, and this forces $D = 3$. The 8-tick epoch defines the fundamental time unit τ_0 and the coherence energy $E_{\text{coh}} = \varphi^{-5}$.

3.9 T8: Three Spatial Dimensions

Proposition 3.9 ($D = 3$ Forced)**.** *The equation $2^D = 8$ (from T7) has the unique positive-integer solution $D = 3$. This is confirmed by two independent structural requirements: (i) nontrivial topological linking (knots exist only in $D = 3$), and (ii) gap-45 synchronization ($\text{lcm}(8, 45) = 360$, yielding angular closure).*

Proof. Primary: $2^D = 8$ has the unique positive-integer solution $D = 3$ (since $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, and 2^D is strictly increasing). Confirmation (i): nontrivial knots require $D = 3$ (in $D \neq 3$, curves are either trivially linked or cannot be knotted). The recognition ledger requires nontrivial linking to maintain closure of the 8-tick walk; hence $D \geq 3$. Combined with $2^D = 8$, this forces $D = 3$. Confirmation (ii): the gauge angle structure has a 45-unit gap (the Gap-45 synchronization); for angular closure, the total angular period must be divisible by both $8\tau_0$ (the tick period) and 45 (the gap); $\text{lcm}(8, 45) = 360$, which is the full circle in degrees—confirming $D = 3$ independently. □

4 Quantum Mechanics Emerges

4.1 Complex Hilbert Space

Proposition 4.1 (Complex Numbers — Structural Argument)**.** *The 8-tick epoch has half-period 4 ticks, corresponding to a phase rotation of π. The RS identification is that representing this requires a number system in which $e^{i\pi} = -1$ has a solution: $\mathbb{C}$.*

Proof. $\mathbb{R}$ cannot represent $e^{i\pi} = -1$ internally (there is no element i with $i^2 = -1$). $\mathbb{H}$ (quaternions) would require the phase structure to accommodate all 8 generators $\{\pm 1, \pm i, \pm j, \pm k\}$ of the quaternion group Q_8, which has 3 independent imaginary units — more structure than the single phase rotation $e^{i\pi}$ of the 8-tick half-period. In the RS framework, the 8-tick complex phase $e^{2\pi ik/8}$ for $k = 0, \ldots, 7$ generates $\mathbb{Z}_8 \subset \mathbb{C}^\times$; this is naturally a subgroup of $\mathbb{C}^\times$, not of the quaternion group. Only $\mathbb{C}$ provides the minimal number system accommodating both the half-period $e^{i\pi} = -1$ and the cyclic 8-tick structure. (This argument is a structural identification rather than a proof of uniqueness; a complete derivation from the RS Hamiltonian is an identified open problem.) □

Corollary 4.2 (Euler's Identity)**.** *$e^{i\pi} + 1 = 0$ is a structural necessity: i is the square root of the half-epoch operator, and Euler's identity records that the half-epoch returns the state to its negative.*

4.2 The Born Rule

Proposition 4.3 (Born Rule from Phase Invariance and Gleason's Theorem)**.** *The probability of a measurement outcome is $P = |A|^2$, where A is the amplitude.*

Proof. The RS recognition rate $R(Ae^{i\theta})$ for a state with amplitude A and phase θ must satisfy:

(i) Phase invariance: $R(Ae^{i\theta}) = R(A)$ (the recognition rate cannot depend on the gauge-choice phase).

(ii) Subadditivity: $R(A + B) \leq R(A) + R(B)$.

(iii) Normalization: $R(1) = 1$.

By (i), R depends only on $|A|$, so $R(A) = f(|A|)$ for some function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$. By (iii), $f(1) = 1$. By Gleason's theorem [4] (which requires dim ≥ 3, ensured by T8), any frame function on a Hilbert space of dimension ≥ 3 must take the form $f(r) = r^2$, i.e., $R(A) = |A|^2$.

Note on the subadditivity condition: The condition (ii) as stated (subadditivity) is weaker than Gleason's actual hypotheses, which require the probability assignment to be *additive* on orthogonal subspaces. The RS structural motivation for additivity is that orthogonal recognition channels are independent and their combined J-cost is additive; subadditivity follows from additivity. The full derivation of the additivity condition from the RS double-entry structure is an identified open problem; here we appeal to Gleason's theorem with the RS confirmation that $D = 3$ ensures dim ≥ 3. □

4.3 Entanglement

Definition 4.4 (Entanglement as Shared Constraint)**.** Two recognition events A and B are entangled when their joint ledger state has a conserved quantity $Q_{AB} = Q_A + Q_B = q_0 = \text{const}$. This forces correlations: knowing Q_A instantly determines $Q_B = q_0 - Q_A$.

Remark 4.5. No action at a distance is required. The correlation is in the ledger entry, not in a signal traveling between A and B. Bell inequality violation occurs because the shared ledger constraint creates a non-factorizable J-cost that no local hidden variable model can reproduce.

4.4 Commutation Relations

Proposition 4.6 (Canonical Commutation — Structural Argument)**.** $[\hat{x}, \hat{p}] = i\hbar$*, where* $\hbar = E_{\text{coh}}\tau_0 = \varphi^{-5}\tau_0$.

Proof sketch. The ledger's double-entry structure (T3) makes the order of observation matter: a position measurement at tick t commits a ledger entry (setting x to a definite value), which back-reacts on the momentum state through the recognition update at $t+1$. The canonical commutation relation arises from the discrete group structure of the ledger updates: the translation operators $\hat{T}_x(\Delta x) = e^{i\Delta x\,\hat{p}/\hbar}$ and $\hat{T}_p(\Delta p) = e^{-i\Delta p\,\hat{x}/\hbar}$ satisfy

$$\hat{T}_x(\Delta x)\hat{T}_p(\Delta p) = e^{i\Delta x\,\Delta p/\hbar}\,\hat{T}_p(\Delta p)\hat{T}_x(\Delta x),$$

giving $[\hat{x}, \hat{p}] = i\hbar$ in the infinitesimal limit. The coherence quantum $\hbar = E_{\text{coh}}\tau_0 = \varphi^{-5}\tau_0$ sets the scale. A complete derivation showing that the RS recognition operator $\widehat{R}$ generates exactly this Weyl–Heisenberg algebra is available in the detailed RS architecture; the above sketch establishes the structural form of the commutator. □

4.5 The Measurement Problem: Dissolved

Proposition 4.7 (Measurement is Recognition — Structural Identification)**.** *In RS, measurement is identified with a recognition event. The "collapse" of the wavefunction is the discrete update of the ledger at the completion of the 8-tick epoch. Under this identification, no additional measurement postulate is needed.*

Proof. The RS ontology distinguishes the ledger (physical reality) from ψ (a computational description of J-cost-weighted probability distributions over ledger states). Superposition exists in ψ, not in the ledger. A recognition event (measurement) produces a new ledger entry, which is either present (event occurred) or absent (did not occur)—no superposition in the ledger. The "collapse" is the updating of ψ to match the observed ledger entry. □

4.6 Decoherence

The decoherence time for a system with energy uncertainty ΔE is

$$\tau_D = \frac{\tau_0}{\Delta E/E_{\text{coh}}} = \frac{E_{\text{coh}}\,\tau_0}{\Delta E} = \frac{\hbar}{\Delta E}, \tag{4}$$

since $\hbar = E_{\text{coh}}\,\tau_0 = \varphi^{-5}\,\tau_0$. For macroscopic objects ($\Delta E \gg E_{\text{coh}}$), $\tau_D \ll \tau_0$—decoherence is effectively instantaneous, consistent with classical behavior.

4.7 Interpretation Resolution

Interpretation	RS Assessment
Copenhagen	Correct operationally: "don't ask what ψ is" works because ψ is a tool, not ontology.
Many-Worlds	Incorrect: there is one ledger, not many branches.
Bohmian	Partially correct: the pilot wave is the J-cost gradient field.
Relational	Correct at the price of circularity: RS provides the non-relational foundation (the ledger).

5 The Quantum-to-Classical Transition

As the number of ledger cells $N \to \infty$, the law of large numbers gives $\langle x \rangle_N \to E[x]$. The key observation is that for macroscopic systems, fluctuations of x_i around their mean become small (relative fluctuation $\sim N^{-1/2}$), so the J-cost average is well approximated by J evaluated at the mean:

$$\langle J \rangle_N = E[J(x)] \approx J(E[x]) + O(N^{-1}). \tag{5}$$

(By Jensen's inequality, $J(E[x]) \leq E[J(x)]$ with equality only when all x_i are equal—i.e., in the classical deterministic limit.) In the large-N limit, individual cells track the mean, and dynamics become deterministic. This is the classical limit: quantum fluctuations of order $\hbar/(N \,\text{action})$ are negligible for macroscopic N.

6 Conclusion

Quantum mechanics is derived from cost minimization. The single primitive—the Recognition Composition Law—forces the unique cost function J, the 8-tick discrete dynamics, and the recognition operator $\widehat{R}$. The results:

1. Complex Hilbert space: forced by the 8-tick half-period $e^{i\pi} = -1$.
2. Born rule $P = |\psi|^2$: from phase invariance, subadditivity, and normalization.
3. Entanglement: from shared ledger constraints.
4. $[\hat{x}, \hat{p}] = i\hbar$: from double-entry non-commutativity, with $\hbar = \varphi^{-5}\tau_0$.
5. Measurement collapse: from the discrete ledger update at $C \geq 1$.

No measurement postulate, no Hilbert-space axiom, and no free parameters. Quantum mechanics is *forced*, not postulated.

References

[1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, New York (1966).

[2] L. Hardy, "Quantum Theory from Five Reasonable Axioms," arXiv: quant-ph/0101012 (2001).

[3] G. Chiribella, G. M. D'Ariano, and P. Perinotti, "Informational Derivation of Quantum Theory," Phys. Rev. A **84**, 012311 (2011).

[4] A. M. Gleason, "Measures on the Closed Subspaces of a Hilbert Space," J. Math. Mech. **6**, 885 (1957).

[5] W. H. Zurek, "Decoherence, Einselection, and the Quantum Origins of the Classical," Rev. Mod. Phys. **75**, 715 (2003).

[6] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). doi:10.3390/axioms15020090